
BrainSegFounder: Towards 3D foundation models for neuroimage segmentation

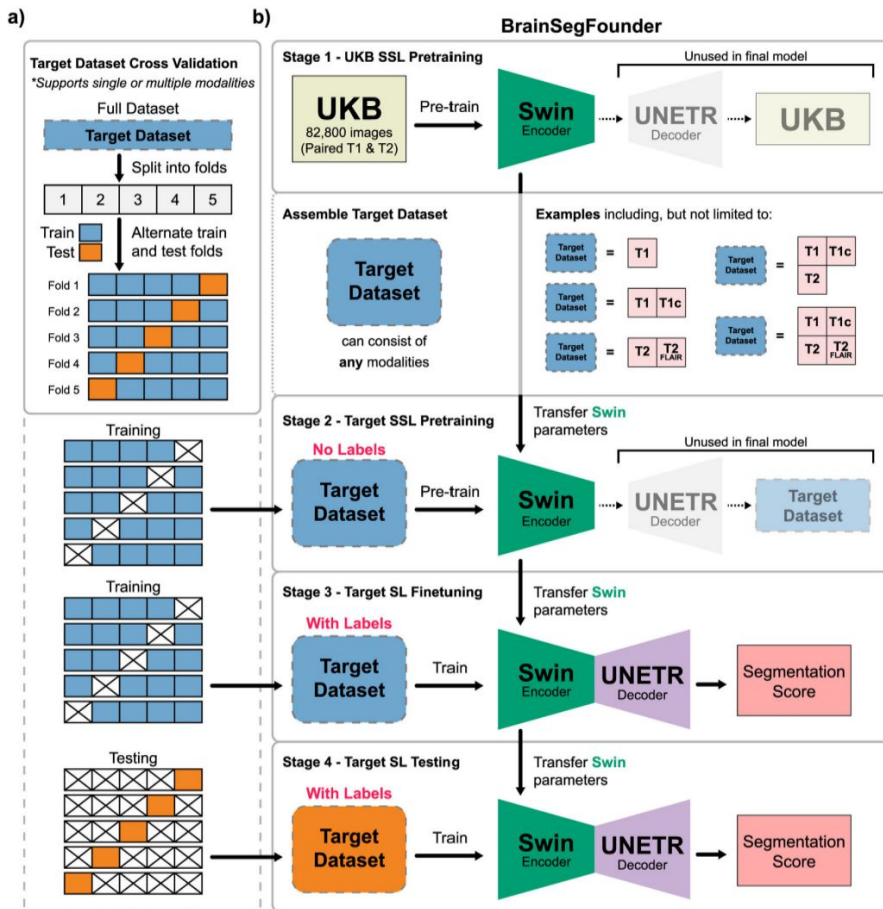
Joseph Coxa, Peng Liua, Skylar E. Stoltea, Yunchao Yangb, Kang Liuc, Kyle B. Seea, Huiwen Jud, Ruogu Fang

Department of Biomedical Engineering, Herbert Wertheim College of Engineering, University of Florida, Gainesville, USA

DOI: 10.1016/j.media.2024.103301

Medical Image Analysis, 2024 October ; Vol. 97

Overall Study Design



Swin = Shifted Window Transformer (windowed self-attention)

Stage 1: Self prelearning on healthy brain on encoder only (UKB dataset) (Masked volume inpainting loss, 3D image rotation loss, contrastive coding loss)

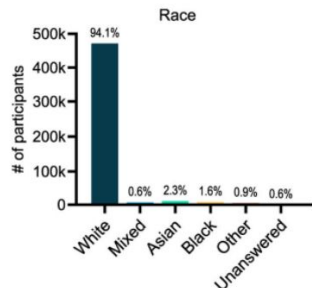
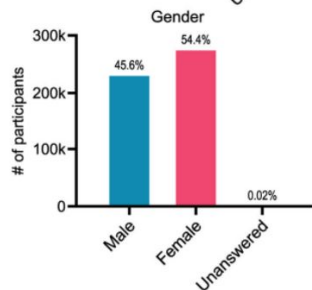
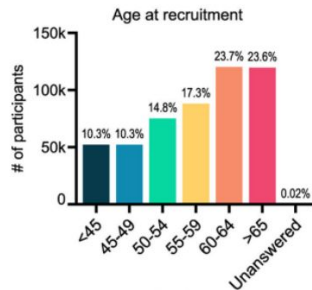
Stage 2: Self prelearning of pathological spatial patterns (BRaTS for tumors, and ATLAS V2 for stroke lesions) (Dice + Cross-Entropy loss)

Stage 3+4: Fine tuning with labels (ground truth)

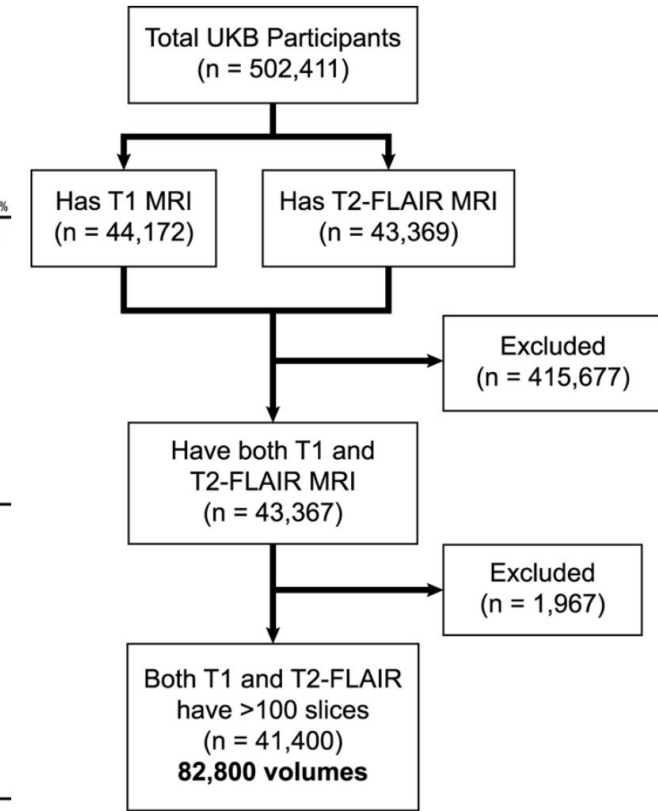
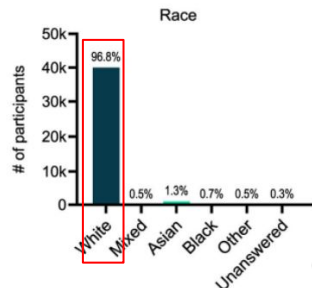
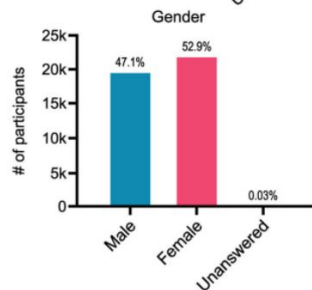
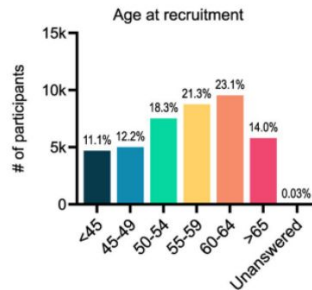
	Data	No. subjects	GPU	Batch size	Learning rate	No. steps	No. input channel
S1	UKB	43,369	64 × A100	128	1×10^{-6}	2×10^5	2
S2+3	BRaTS	1251	2 × A100	2	1×10^{-4}	50,000	4
	ATLAS v2.0	655	4 × A100	4	3×10^{-3}	600	1

Datasets: UK biobank, BraTS, ATLAS

Total UK Biobank



40k Cohort



CONSORT diagram of UK biobank

UK Biobank

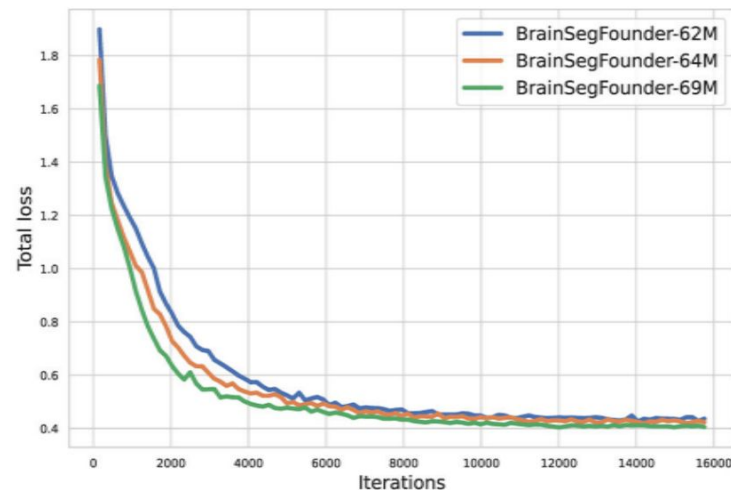
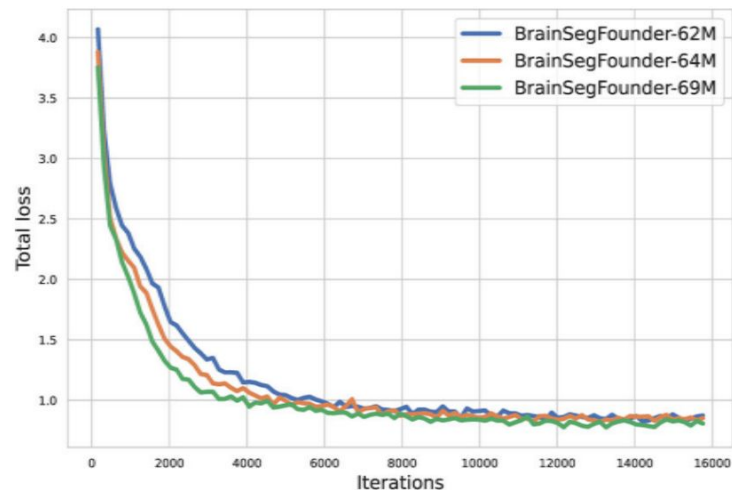
Number of subjects	41,400
Modalities	T1w, T2-FLAIR
Number of images	82,800
Diseases	Generally healthy

**For stage 1
(Pretraining for encoder only)**

BraTS	ATLAS
1251	655
T1w, T1-ce, T2w, T2-FLAIR	T1-ce
5004	655
Malignant brain neoplasms	Stroke

**For stage 2+3
(Pretraining & Fine-tuning)**

Impact of scaling

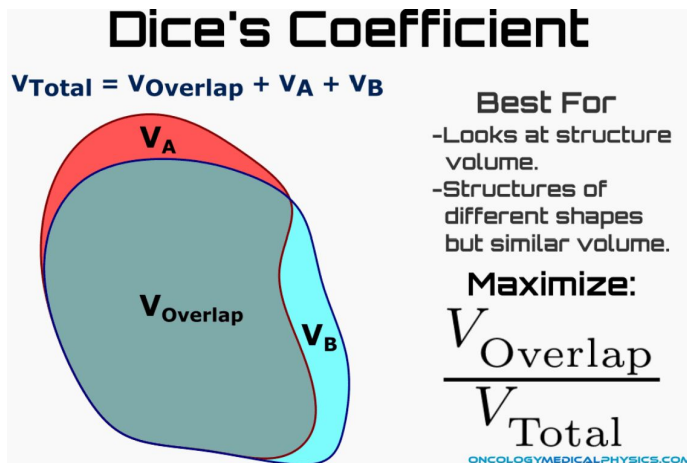


# of encoder parameters		BrainSegFounder-Tiny (62M)		BrainSegFounder-Small (64M)		BrainSegFounder-Big (69M)	
		19,097,191		20,982,103		26,636,839	
Encoder layer level	Output size	# of SSL heads	# Swin blocks	# of SSL heads	# Swin blocks	# of SSL heads	# Swin blocks
Level 1	$48 \times (48 \times 48 \times 48)$	3	2	3	2	3	2
Level 2	$96 \times (24 \times 24 \times 24)$	6	2	6	2	6	2
Level 3	$192 \times (12 \times 12 \times 12)$	12	2	12	6	12	18
Level 4	$384 \times (6 \times 6 \times 6)$	24	2	24	2	24	2
Feature size		48		48		48	
Bottleneck dimension		768		768		768	

Training (left) and validation (right) loss of Stage 1-pretraining three different scale of BrainSegFounder models on UKB (different encoder settings: how many sliding-window blocks)

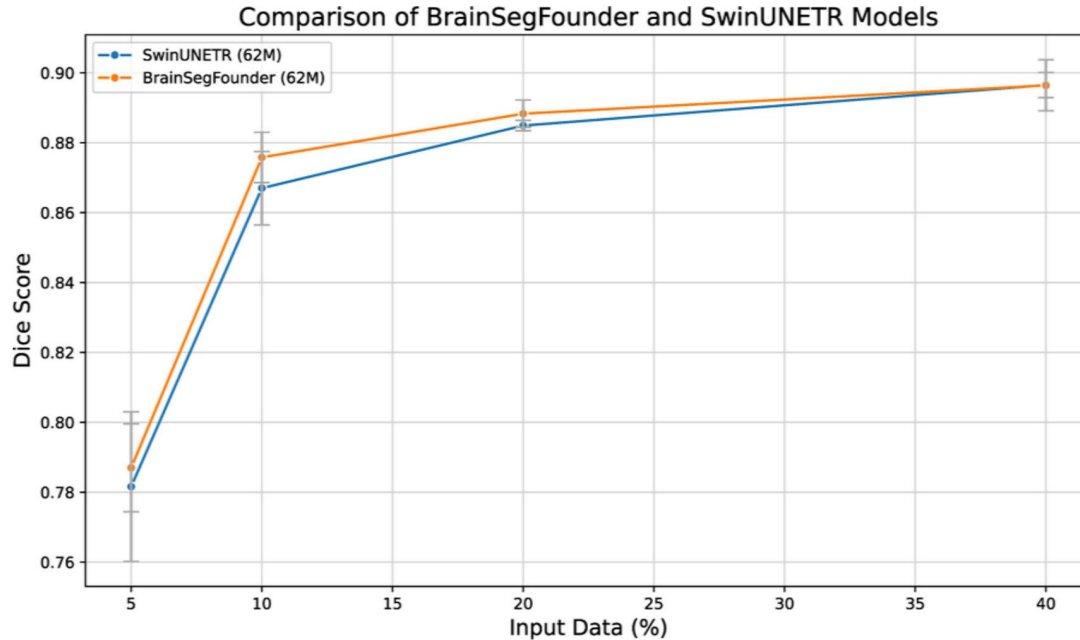
Result comparison: Average dice coefficient

	BrainSegFounder (T1w only)	SwinUNETR (T1w only)
Fold 1	0.718	0.700
Fold 2	0.721	0.672
Fold 3	0.707	0.657
Fold 4	0.731	0.686
Fold 5	0.725	0.676
Average	0.721	0.678



SwinUNETR (baseline) is fully supervised learning on T1-weighted MRI without pretraining, while **BrainSegFounder** uses our multi-stage pretraining on UKB and BraTS T1-weighted MRI and is then finetuned on BraTS T1-weighted MRI.

Impact of data availability



Dice coefficients for baseline (SwinUNETR) and our model across different levels of training data availability. All models were trained 5 times to account for variability in the input data randomly selected. Error bars represent $\pm$ one standard deviation.

BraTS challenge: Average dice coefficient

	BSF-S (64M)	BSF-1-S (64M)	SwinU-MRI	SwinU-Res	SwinU-CT	nnU-Net	SegResNet	TransBTS	Model-Zoo
Fold 1	0.9032	0.8994	0.8854	0.895	0.894	0.896	0.899	0.883	0.857
Fold 2	0.9182	0.9055	0.9059	0.899	0.902	0.917	0.916	0.902	0.879
Fold 3	0.9121	0.9125	0.8981	0.894	0.898	0.910	0.909	0.889	0.820
Fold 4	0.9100	0.9133	0.8924	0.890	0.893	0.909	0.908	0.893	0.889
Fold 5	0.9139	0.9114	0.9035	0.903	0.902	0.909	0.906	0.892	0.893
Average	0.9115	0.9110	0.8971	0.896	0.898	0.908	0.907	0.891	0.868

BSF-1-S indicates this model with only the Stage 1 (SSL) pertaining on UKB and without the Stage 2 pretraining step.

SwinU models are models using the SwinUNETR architecture trained on BraTS via supervised learning.

SwinU-MRI is the model trained directly using supervised learning on BraTS published on GitHub

SwinU-Res is pretrained with SSL on only T1w and T2w and finetuned on BraTS

SwinU-CT pretrained using CT data and finetuned with supervised learning on BraTS.

nnU-Net and **SegResNet** are former BraTS challenge winners trained using supervised learning on our folds.

TransBTS is a vision-transformer based segmentation algorithm optimized for brain segmentation.

Model-Zoo is a bundle of models published by MONAI that can perform BraTS segmentation out of the box using their

Brats mri segmentation

Results comparison: ATLAS v2.0 dataset

Metric	BrainSegFounder	SwinUNETR	CTRL([*] †)	HeRN([*] ‡)	POBOTRI([*])
Dice (↑)	0.712	0.703	0.663	0.718	0.663
Lesion-wise F1 Score (↑)	0.711	0.703	0.556	0.724	0.559
Simple Lesion Count (↓)	3.421	3.677	4.657	2.750	4.500
Volume Difference (↓)	8993.85	9165.18	8804.91	6162.00	9535.23

Scores for models marked with an asterisk (*) are sourced directly from the official challenge leaderboard and pertain to their performance on the training set. All of these models utilize ensemble learning methods. CTRL (denoted with †) is the official challenge winner, while HeRN (denoted with ‡) leads on the training set. Top-performing models on the leaderboard (CTRL, HeRN, and POBOTRI) were unavailable for independent validation.

Discussion

Results conclusion: Swin encoder is well-suited for 3D medical images because it captures both fine local structure (individual brain regions) and large-scale global context (spatial layout of the whole brain) efficiently, without requiring labels during pretraining

Few-shot learning: BrainSegFounder consistently outperformed SwinUNETR when training data was limited (5-20% of BraTS data), demonstrating better generalization under data scarcity.

Modality flexibility: The model adapts to fewer or more input modalities without architectural changes, confirmed by ablation showing a 6% Dice improvement over SwinUNETR using only T1w MRI.

Advantages over SAM/MedSAM: Operates natively in 3D, requires no manual prompts at inference, handles multi-modal inputs, and produces a fixed number of segmentation classes.

Limitations: The 40K UK Biobank cohort is 96.8% White, raising generalizability concerns. Brain tissue segmentation and other neurological tasks beyond tumors and stroke remain unexplored. Larger models didn't consistently improve results, suggesting dataset scale matters more than parameter count alone.

Thanks for your attention !